HealthX: LOW-COST IOT-BASED HEALTH MONITORING SYSTEM FOR REMOTE COMMUNITIES

A study presented to the faculty of

School of Engineering, Computer, and Library Studies

Holy Cross College

Sta. Lucia, Sta. Ana, Pampanga

In Partial Fulfillment of the Requirements for the Degree

Bachelor of Science in Computer Engineering

Submitted by:

CAYABYAB, Matt Julius M.

DEANG, Ronnel O.

DIGMAN, Christian D.

PARAGAS, John Ian Joseph M.

NOVEMBER 2025

CHAPTER I

INTRODUCTION

Health care is one of the most vital foundations of any community, yet many remote areas continue to struggle with limited access to medical services. Residents often face barriers such as distance from hospitals, the high cost of regular checkups, and the shortage of available doctors. These challenges frequently result in delays in the detection of illnesses, which can pose serious health risks to individuals and families. With the advancement of technology, particularly through the Internet of Things (IoT), new opportunities are emerging to bridge the gap between healthcare providers and underserved communities.

This study introduces HealthX: A Low-Cost IoT-Based Health Monitoring System for Remote Communities, developed to address the gaps in existing healthcare monitoring technologies and provide a more accessible, community-centered alternative. Unlike the RxBox, the proposed system emphasizes affordability, portability, and real-time monitoring through the use of an ESP32 microcontroller integrated with multiple sensors. It monitors essential health parameters including oxygen saturation (SpO_2), heart rate, body temperature, blood pressure, and body mass index (BMI), thereby offering a more comprehensive range of health indicators compared to the

RxBox. These parameters were specifically selected because they represent the most fundamental indicators of an individual's physiological condition. Monitoring body temperature, heart rate, SpO_2, and blood pressure provides immediate insights into cardiovascular, respiratory, and metabolic health, while BMI serves as an important metric for assessing nutritional status and long-term health risks. Collectively, these metrics allow for early detection of abnormalities and the promotion of preventive healthcare, making them highly relevant for continuous community-based monitoring.

According to the World Health Organization (2020), the normal human body temperature typically ranges from 36.5°C to 37.5°C, while the American Heart Association (2021) reports that a normal resting heart rate for adults falls between 60 and 100 beats per minute. Oxygen saturation (SpO_2) levels are considered healthy when they range from 95% to 100%, as noted by the Centers for Disease Control and Prevention (2021), and normal blood pressure is approximately 120/80 mmHg (American Heart Association, 2021). Additionally, the World Health Organization (2019) defines a healthy body mass index (BMI) as ranging from 18.5 to 24.9.

Despite the availability of these established health standards, consistent monitoring of vital signs remains a major challenge, particularly in rural and low-resource communities where access to medical facilities and professional healthcare services is limited. Many existing monitoring devices are costly, function as standalone units, and lack the integration required for a holistic health assessment. To address these challenges, the HealthX system aims to provide an affordable, efficient, and multi-parameter IoT-based solution capable of continuously tracking key health indicators—body temperature, heart rate, SpO_2, blood pressure, and BMI—to support accessible and proactive healthcare monitoring in underserved areas.

Previous studies have explored the potential of IoT-based systems in addressing similar challenges. Patel et al. (2022) developed an IoT-based patient monitoring device utilizing the ESP32 microcontroller for wireless data transmission, proving its efficiency, low cost, and reliability in real-time health data collection. Likewise, Gupta and Kumar (2021) integrated the MAX30102 sensor for heart rate and SpO_2 monitoring, confirming its accuracy and suitability for portable health devices. Rahman et al. (2020) employed the DS18B20 temperature sensor in a patient health monitoring system, demonstrating its precision in body temperature detection. In addition, Singh et al. (2021) incorporated a load cell with an HX711 amplifier

to measure body weight and compute BMI, reinforcing the sensor's capability in health-related applications. These studies collectively support the effectiveness of using the ESP32, MAX30102, DS18B20, and HX711 modules in developing reliable and affordable health monitoring systems—an approach that serves as the foundation for the proposed HealthX system.

The system operates through an integrated workflow that combines hardware and software components to facilitate real-time health monitoring. It utilizes an ESP32 Wi-Fi IoT microcontroller as the central processor and wireless hotspot, a MAX30102 sensor for measuring blood oxygen saturation (SpO_2) and heart rate, a DS18B20 digital sensor for body temperature measurement, a load cell with an HX711 amplifier for weight monitoring, and a blood pressure sensor module. Programmed using C++ in the Arduino IDE, the ESP32 collects and processes data from each sensor while establishing a local Wi-Fi network that enables smartphone connection. The Flutter-based mobile application is designed for two user roles: the admin (health worker) and the user (patient). The admin interface allows health workers to access live calibration readings and monitor patients' real-time data during testing, while the user interface is limited to viewing recorded health data and past results. After conducting tests, the health worker transmits the collected data to a PHP/MySQL web dashboard, where healthcare professionals can review

information, analyze historical trends, and provide timely medical guidance. This integrated approach enhances accuracy, efficiency, and accessibility in healthcare delivery, particularly for patients in remote or underserved areas.

Background of the Study

Health care access is a growing concern in many rural areas, particularly in municipalities like Sta. Ana, Pampanga, where residents face limited access to hospitals, diagnostic equipment, and professional medical care. Communities in these areas often rely on small health centers with limited staff and outdated tools, making regular check-ups and preventive health care difficult. Patients with chronic illnesses such as hypertension or respiratory problems are at risk of late diagnosis or untreated conditions, which can lead to serious complications. The distance to urban hospitals, combined with financial limitations, further reduces opportunities for timely medical intervention.

The HealthX system aims to address these challenges by providing an affordable, accessible, and real-time way for individuals to track their vital health information. The system uses an ESP32 microcontroller as its main processor, connected to a set of sensors that measure oxygen saturation (SpO_2), heart rate, body temperature, blood pressure, and body mass index (BMI). These vital signs were carefully selected because they serve as

primary indicators of an individual's overall physiological condition and are widely recognized in clinical practice as essential for assessing cardiovascular, respiratory, and metabolic health. Monitoring body temperature allows the early detection of fever and infection; heart rate provides insight into cardiac activity and stress levels; SpO_2 reflects oxygen delivery efficiency and respiratory function; blood pressure indicates circulatory health and the risk of hypertension; and BMI offers a measure of nutritional status and long-term disease risk such as obesity or malnutrition. By integrating these parameters, the HealthX system enables continuous and comprehensive monitoring of a person's health, facilitating early detection of potential conditions such as fever, hypertension, and hypoxemia. This selection ensures that the system remains both clinically relevant and practical for deployment in rural and underserved communities, where access to professional healthcare and diagnostic tools is often limited.

Several studies have demonstrated the potential of IoT-based health monitoring systems using similar hardware configurations. Patel et al. (2022) developed a patient monitoring system that utilized the ESP32 microcontroller for real-time data acquisition and wireless transmission, proving its cost-effectiveness and efficiency in remote healthcare applications. Similarly, Gupta and Kumar (2021) employed the MAX30102

sensor to accurately measure pulse rate and blood oxygen levels, confirming its suitability for non-invasive monitoring. Rahman et al. (2020) integrated the DS18B20 temperature sensor with IoT technology to monitor body temperature, achieving high accuracy and reliability in continuous data tracking. Additionally, Singh et al. (2021) implemented a load cell with an HX711 amplifier to calculate BMI as part of a low-cost health assessment system. These related works collectively support the practicality of using the same components and IoT framework in developing an accessible health monitoring solution such as HealthX.

The device is designed with both user and healthcare provider needs in mind. On the user side, the system includes a mobile application that records health data, stores history, and provides alerts whenever abnormal readings are detected. For instance, if the body temperature drops below or rises above the normal range, the app notifies the user and provides guidance on possible actions. On the healthcare provider side, a web-based dashboard enables doctors to monitor patients remotely, allowing them to review real-time readings and historical data. This dual interface ensures that patients maintain continuous access to their health information while healthcare professionals can provide oversight and medical advice even without in-person consultations.

The system operates through a straightforward process: sensors collect the user's vital signs, the ESP32 processes the data, the information is fetched by the user's smartphone, and the results are displayed through a Flutter-based mobile application. The collected data is then stored and updated on a PHP/MySQL web dashboard, ensuring that medical records remain current and accessible. This continuous data cycle supports proactive health monitoring, enabling early detection and timely medical response.

Such an approach aligns with the World Health Organization's (WHO, 2022) emphasis on leveraging digital health technologies to strengthen primary healthcare services in underserved communities. In addition, Ahmed and Rahman (2021) and Patel et al. (2019) emphasized that IoT-based health monitoring systems significantly improve early detection of health risks, reduce healthcare costs, and enhance accessibility in rural populations. Their findings demonstrate the transformative potential of integrating IoT technologies into community health systems, particularly in regions with limited medical infrastructure.

By focusing on the communities in Sta. Ana, Pampanga, this study not only adapts these proven concepts within a local context but also goes beyond existing solutions such as the RxBox. The HealthX system is designed to be more affordable, flexible, and comprehensive in terms of the health indicators it monitors. This innovation empowers residents to self-monitor their health while maintaining a direct connection with healthcare professionals. In doing so, it reduces medical costs, saves time, and strengthens preventive care—ultimately improving the overall well-being of underserved communities.

Review of Related Literature & Studies

Foreign Literature

Rahman et al. (2020) presented the "**IoT Based Health Monitoring System for Patients**" which utilized a Raspberry Pi with heart rate and SpO_2 sensors for remote patient tracking. Their system highlighted the potential of IoT in medical monitoring but focused only on limited parameters. The present study extends this work by applying an ESP32 microcontroller to measure six vital indicators oxygen saturation, heart rate, body temperature, blood pressure and BMI making it more comprehensive and practical for underserved communities.

Albahri et al. (2021) proposed "**RealTime Health Monitoring Using IoT and Cloud Architecture**" which measured oxygen saturation and heart rate while transmitting data wirelessly. Their research emphasized emergency detection for cardiac issues. In contrast, the current study advances this approach by integrating additional health metrics such as BMI, ensuring continuous monitoring of multiple conditions relevant to rural populations with limited hospital access.

Rathore and Gupta (2022) examined "**Smart IoT Based Wearable Health Devices**" designed for fitness monitoring, calorie intake, and vital sign tracking. These devices were aimed at urban populations with lifestyle goals. Unlike their work, the present research applies ESP32 technology to engineer a low-cost, multi-parameter health monitoring system designed specifically for rural communities, prioritizing accessibility and preventive diagnostics over fitness tracking.

Local Literature

Estrella et al. (2019) designed "**IoT Based Remote Health Monitoring for the Philippines**" which measured blood pressure and pulse rate for elderly patients using GSM communication. Their system assisted caregivers but lacked wider diagnostic functions. The present study builds on this by implementing The ESP32 microcontrollers creates a hotspot for the user's phone to capture multiple health parameters temperature, SpO_2, BMI, glucose, and more providing a more holistic health solution for Filipino families in remote barangays.

Rodriguez and Dizon (2020) implemented "**Affordable IoT Enabled Health Monitoring for Low Income Communities**" which focused on blood pressure and glucose monitoring at reduced cost. While effective, it covered only two parameters. The proposed study enhances this by consolidating six health indicators in a single device, strengthening early detection of hypertension, and respiratory issues while keeping affordability central for rural households.

The Department of Science and Technology (2016) engineered "**RxBox**", a telehealth device capable of six diagnostic tests including temperature, SpO_2, blood pressure, and ECG. It was designed for remote areas but remains expensive and dependent on institutional distribution. The present study follows the same advocacy for improved healthcare access but distinguishes itself by applying ESP32 to construct a low-cost, community-level monitoring device affordable for households without government intervention.

Foreign Studies

Patel and Sharma (2020) analyzed the "**IoT Based Wireless Patient Health Monitoring System**" which alerted doctors via Wi-Fi in emergency cases. While their design focused on rapid response, the present study emphasizes preventive monitoring of six vital signs to detect risks early, reducing emergency cases in underserved populations.

Ahmed et al. (2021) investigated the "**Cloud Based IoT Health Monitoring for Blood Pressure**" which primarily tracked glucose levels and suggested lifestyle adjustments. Although beneficial for blood pressure care, it was limited to one condition. The present study expands this by measuring glucose along with other vital signs such as blood pressure and oxygen saturation, addressing multiple chronic illnesses common in rural settings.

Chen and Li (2022) applied IoT in their "**Wearable ECG and Heart Rate Monitoring System**" tailored for hospital-based care. Their framework relied on existing medical infrastructures. By contrast, the present research develops a system intended for household use in rural communities, enabling continuous health tracking without advanced hospital facilities.

Local Studies

Santiago (2018) proposed "**IoT Based Blood Pressure and Heart Rate Monitoring Device for Barangay Health Centers**" to assist rural clinics with limited diagnostic tools. However, it was restricted to two parameters. The current research improves this by offering six measurements in a single device, giving barangays health workers broader diagnostic capacity for community health monitoring.

Mendoza and Cruz (2019) constructed "**Community Health Monitoring Using Arduino-Based Systems**" to provide affordable tools for clinics and schools. Their design achieved cost-effectiveness but lacked wireless data transmission. The present study addresses this gap by utilizing ESP32 with Wi-Fi connectivity, enabling real-time patient data sharing for quicker healthcare response.

The University of the Philippines Diliman and the Department of Science and Technology (2016) developed the **RxBox**, a telemedicine device that transmits patient data from rural areas to specialists. While effective in expanding healthcare access, it remains costly, dependent on government programs, and limited in monitored parameters. This study addresses those gaps with a low-cost, portable IoT-based system that tracks a wider range of vital signs in real time, offering an affordable, scalable solution that rural communities can operate independently.

Conceptual Framework

Figure 1

Input–Process–Output (IPO) Model of the HealthX System

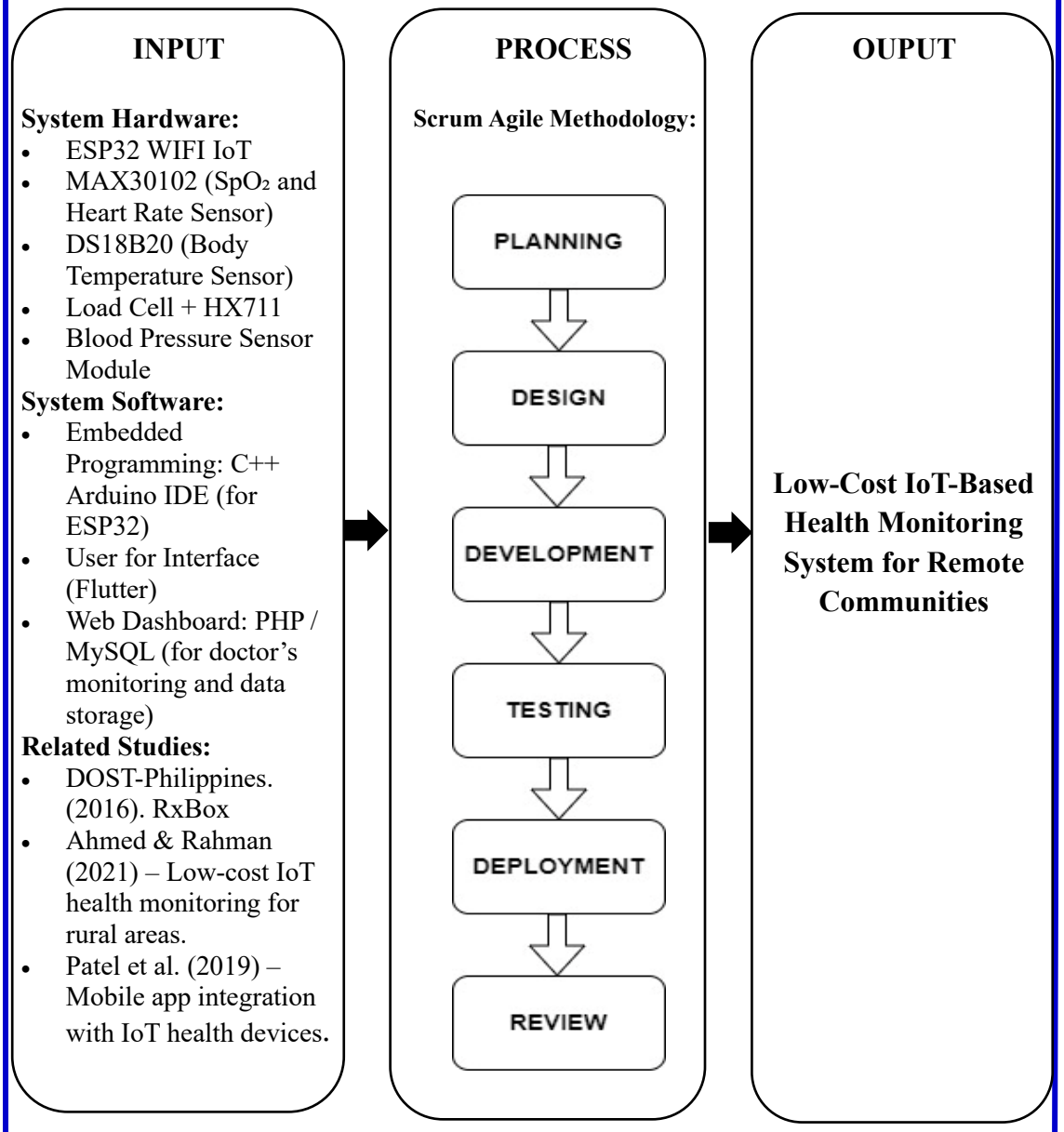

Figure 1 shows that creation of a Low-Cost IoT Based Health Monitoring System for Remote Communities. The HealthX system integrates both hardware and software components to deliver an intelligent, real-time health monitoring solution that promotes accessible and affordable healthcare. The hardware inputs include the Wi-Fi IoT board (ESP32), MAX30102 sensor, DS18B20 temperature sensor, load cell with HX711 amplifier, and a blood pressure sensor, all of which work cohesively to collect and process vital health data. The ESP32 functions as the central controller, facilitating sensor communication, data processing, and wireless transmission to the web and mobile platforms. The MAX30102 measures blood oxygen saturation (SpO_2) and heart rate through optical sensing, while the DS18B20 captures accurate body temperature readings. The load cell, paired with the HX711 amplifier, measures weight to compute the Body Mass Index (BMI), and the blood pressure sensor detects systolic and diastolic values essential for cardiovascular assessment. The software aspect complements the hardware through C++ programming in the Arduino IDE for ESP32 control, a Flutter mobile app for real-time data visualization, and a PHP/MySQL web dashboard that allows healthcare professionals to monitor patient data remotely. Guided by the principles of the Scrum Agile methodology and inspired by initiatives such as the DOST-Philippines RxBox (2016), the

system's development followed a structured yet flexible process consisting of planning, design, development, testing, deployment, and review phases. Each stage emphasized collaboration, adaptability, and continuous improvement to ensure the reliability and usability of the system. Ultimately, the integration of these technologies resulted in a real-time monitoring platform capable of recording and displaying vital signs efficiently, empowering patients and healthcare professionals with data-driven insights for timely diagnosis, proactive care, and improved health outcomes, particularly in underserved rural communities.

Synthesis

The reviewed literature collectively underscores the transformative potential of IoT-based health monitoring systems in enhancing healthcare accessibility, while also exposing critical gaps that the present study seeks to address. Previous investigations have demonstrated the feasibility of real-time physiological monitoring using microcontrollers and wireless communication; however, many were limited to tracking a narrow range of health parameters and relied heavily on hospital-based infrastructures. Other innovations emphasized fitness applications tailored to urban populations, which do not adequately cater to the healthcare needs of rural or underserved

communities. Local initiatives have shown progress in creating affordable and community-oriented monitoring systems, yet these often-lacked diagnostic depth or the capability for continuous wireless reporting. Moreover, existing government-developed solutions remain costly and constrained by centralized implementation. The present study bridges these limitations by developing a low-cost, ESP32-based IoT health monitoring system capable of measuring multiple physiological parameters—oxygen saturation, heart rate, body temperature, blood pressure, and body mass index—in real time. This integration of affordability, comprehensive diagnostics, and scalability at the community level offers a practical and inclusive approach to continuous health monitoring, thereby advancing the goals of accessible and technology-driven healthcare.

Statement of the Problem

Traditional health monitoring in Sta. Ana, Pampanga depends on manual check-ups and infrequent clinic visits. Limited staff, outdated equipment, and distant hospitals cause delayed illness detection and treatment, raising risks for chronic patients. Current monitoring systems are costly and out of reach for many low-income families, hindering continuous care.

This study presents the design and development of a HealthX: Low-Cost IoT-Based Health Monitoring System for Remote Communities. Specifically, it seeks to address the following key questions:

1. How can the system's sensor integration and calibration be optimized to maintain accurate measurements of vital health parameters under varying environmental conditions?
2. How can the system be designed to measure vital parameters such as SpO_2, heart rate, body temperature, blood pressure, and automatic BMI?
3. How can the accuracy and performance of the HealthX system's measured parameters be validated through comparative analysis with existing medical test kits and devices, such as digital thermometers, weighing scales, and standard instruments for blood pressure, SpO_2, and heart rate measurement?
4. How can the system ensure continuous and reliable health data monitoring using an ESP32 microcontroller that fetches data in remote areas like Sta. Ana, Pampanga?
5. How can the integration of a mobile application and a web-based dashboard provide real-time access to health records for both users and healthcare professionals?

6. How can the system be evaluated for software quality using ISO/IEC 25010 and for hardware/data quality using ISO/IEC 25012 to ensure effectiveness, reliability, and usability in real-world deployment?

Significance of the Study

- **To the Users:** Provides affordable, real-time monitoring of vital signs to help patients detect health issues early and manage chronic conditions.

- **To the Community:** Promotes accessible healthcare and reduces risks of late detection in remote areas.

- **To Healthcare Professionals:** Enables remote access to patient data through a web dashboard for timely diagnosis and intervention.

- **To the Researchers:** Enhances the researchers' technical and analytical skills by applying theoretical knowledge of IoT, programming, and system integration to a real-world healthcare application, while contributing to the development of low-cost, ESP32-based sensor technologies.

- **To Future Researchers:** Serves as a reference and guide for future related studies and improvements.

Scope and Delimitations

This study focuses on the development of a low-cost IoT-based health monitoring system designed to assist remote communities with limited access to healthcare facilities. The system integrates multiple health parameters, including SpO_2 (oxygen saturation), heart rate, body temperature, blood pressure, and automatic BMI calculation, to provide a comprehensive health assessment at a fraction of the cost of commercial devices. The research emphasizes the use of IoT-based sensors connected to an ESP32 microcontroller, enabling real-time monitoring and data transmission to a mobile application for user accessibility.

The scope of this study includes:

System Design – The device incorporates non-invasive biomedical sensors to accurately measure vital signs. The ESP32 microcontroller processes the data, which is then fetched by the user's mobile application for real-time display. The design emphasizes affordability, portability, and energy efficiency to ensure effective use in resource-limited areas.

Health Parameters – The system measures five key health indicators—oxygen saturation (SpO_2), heart rate, body temperature, blood pressure, and body mass index (BMI). Normal reference ranges are 95–100% for SpO_2, 60–100 bpm for heart rate, 36.5–37.5°C for body temperature, 120/80 mmHg for blood pressure, and 18.5–24.9 for BMI. These values serve as benchmarks for detecting early signs of conditions such as hypertension, hypoxia, and fever.

IoT Integration – A Flutter-based mobile application is integrated with the system to provide users with easy access to their health data. The app displays results instantly, provides a history log for tracking, and sends alerts when values fall outside the normal range. This improves health awareness and enables better decision-making in communities with limited medical support.

Usability – The system is designed to be user-friendly, requiring minimal technical knowledge. It is intended for use by community health workers, local caregivers, or individuals themselves, making it suitable for remote, rural, and underserved populations.

The delimitations of the study include:

Limited Parameters – While the system measures six health indicators, it does not include advanced diagnostics such as ECG, cholesterol levels, or respiratory rate. These are beyond the scope of this low-cost prototype.

Target Communities – The system is tested for use in remote rural communities with limited healthcare access. It will not initially cover urban hospitals or advanced clinical settings.

Accuracy and Calibration – The system uses low-cost sensors that, while not as precise as medical-grade devices, produce results closely aligned with clinical standards when properly calibrated during the study.

Time frame – The research evaluates the system within a specific testing period to validate accuracy and usability. Long-term durability, seasonal variations, and extended field deployment are not part of this study.

Sample Size – The prototype will be tested on a limited number of volunteers for validation purposes. Large-scale clinical trials are outside the scope of this research.

Definition of Terms

Blood Pressure Monitoring – A feature that measures systolic and diastolic pressure in real time. Normal blood pressure for adults is around 120/80 mmHg. This function helps detect hypertension (high blood pressure) or hypotension (low blood pressure), which are critical indicators of cardiovascular health.

Body Mass Index (BMI) Automation – A function that automatically calculates BMI using weight and height inputs. A normal BMI is between 18.5 and 24.9. This provides a quick assessment of whether an individual is underweight, normal, overweight, or obese, aiding in preventive healthcare.

Body Temperature Monitoring – The process of recording an individual's body heat through a sensor. The normal body temperature is approximately 36.5°C to 37.5°C. Deviations from this range may indicate fever, infection, hypothermia, or other health concerns.

ESP32 WIFI IoT Development Board – The main microcontroller of the system. It gathers data from sensors, processes health parameters, and transmits information wireless to the mobile application for real-time monitoring.

Flutter Mobile Application – The mobile interface that displays health data to users and healthcare workers. It provides easy access to records and continuous health updates for remote monitoring.

Heart Rate Monitoring – A function that detects and records the number of heart beats per minute. The normal resting heart rate for adults is 60–100 beats per minute (bpm). Abnormal readings may indicate tachycardia, bradycardia, or cardiovascular issues.

Oxygen Saturation (SpO_2) Monitoring – A feature that measures the percentage of oxygen in the blood. The normal SpO_2 level is 95%–100%. Levels below 94% may indicate respiratory issues or poor oxygen circulation.

Remote Communities – The target areas of this study where healthcare access is limited. The system offers affordable, continuous, and portable health monitoring solutions to people in underserved regions.

CHAPTER II

METHODOLOGY

The chapter was intended to evaluate the varied methodologies that the researcher could employ to conduct the research study. As important as anything else in the whole research were the different methodologies concerning research design, population, research instruments, data collection processes, and statistical treatment of the data. The importance of these had the utmost say about the accuracy, reliability, and validity of the research findings.

Research Design

The study adopted a quantitative experimental research design to assess the accuracy and reliability of the HealthX system. This approach involved collecting numerical data under controlled conditions to ensure objective and evidence-based results. The controlled variables included environmental factors such as room temperature, participant posture during measurement, and the time interval between tests to maintain consistency across trials. The uncontrolled variables were individual physiological differences among participants, such as stress levels, hydration status, and recent physical activity, which could influence readings. The experiment measured six physiological parameters—oxygen saturation (SpO_2), heart

rate, body temperature, blood pressure and body mass index (BMI)—using the HealthX device and compared them with certified medical instruments. Standard clinical benchmarks, such as normal blood pressure below 120/80 mmHg, served as reference points for evaluating measurement accuracy.

All collected data were subjected to statistical analysis to evaluate consistency, identify deviations, and confirm measurement validity. This process aligns with Creswell and Creswell's (2020) characterization of quantitative research as a structured approach focused on numerical measurement and statistical interpretation. By testing the device in controlled experiments and analyzing results through statistical methods, the study ensured that observed outcomes reflected the true performance of HealthX rather than external variables. The findings thus offer reliable evidence of the system's clinical applicability and inform potential improvements in its design and deployment for community health monitoring.

Development Phases:

1. Planning Phase

The planning phase began with an extensive review of literature on remote health monitoring technologies, IoT-based medical devices, and relevant software standards to establish the specifications for the HealthX. Essential requirements were defined to ensure affordability, portability, and real-time data transmission for rural communities. The hardware design centers on an ESP32 development board as the main controller, chosen for its low cost and integrated wireless capability. Key sensors include the MAX30102 for continuous SpO_2 and heart-rate measurement, the DS18B20 for precise body temperature monitoring, a Load Cell with HX711 module for weight acquisition to calculate Body Mass Index (BMI), and a Blood Pressure Sensor Module for systolic and diastolic readings. Software requirements specify C++ using the Arduino IDE for embedded programming of the ESP32, a Flutter-based mobile application to display real-time vitals and alerts, and a PHP/MySQL web dashboard to allow healthcare professionals to view and store patient data securely. Baseline physiological standards guided all planning: normal human body temperature of 36.5–37.5 °C, heart rate of 60–100 bpm, SpO_2 levels of 95–100 %, blood pressure near 120/80 mmHg.

2. Design Phase

All design exercises focus on creating a detailed circuit diagram and system architecture for the prototype. The ESP32 is configured as the central controller, receiving continuous data from the MAX30102, DS18B20, Load Cell, and blood pressure sensor. The design ensures stable power regulation, noise-free sensor signals, and efficient data flow to the mobile app and web dashboard. User interface mock-ups emphasize clear visualization of each vital sign and instant alerts for readings outside the normal range. Database structures and security measures are documented to protect sensitive medical data.

3. Development Phase

The prototype assembly integrates all sensors with the ESP32 on a single board, with firmware programmed in Arduino C++ to read, filter, and fetched data. The Flutter mobile application is built to display live graphs, historical records, and automated notifications when parameters exceed normal values. Meanwhile, the PHP/MySQL web dashboard is developed to allow healthcare professionals secure access to real-time and archived patient data. Iterative coding and hardware tuning ensure that each sensor maintains medical-grade accuracy, such as ±0.3 °C for temperature and ±2 % for SpO_2.

4. Testing Phase

Testing is conducted following ISO/IEC 25010 for software quality and ISO/IEC 25012 for hardware reliability. Sensor outputs are benchmarked against certified medical devices thermometers, pulse oximeters, sphygmomanometers, and glucometers to ensure accuracy within standard clinical tolerances. Connectivity tests assess the reliability of data fetching under real rural conditions, including latency and data integrity. Power-consumption measurements verify that the system remains affordable and energy-efficient for households with limited electricity access. Additionally, pilot testing is conducted in the target community to evaluate the system's usability, reliability, and overall performance in real-world conditions.

5. Evaluation Phase

The final evaluation was conducted at a community health center in Sta. Ana, Pampanga, where data collected over several weeks were analyzed using mean, standard deviation, and regression analysis to assess accuracy, reliability, and user satisfaction. Environmental factors, such as ambient temperature, were also examined for their effect on sensor performance, and feedback from healthcare providers and patients guided recommendations for system enhancement and wider use.

Respondents

The residents of Sta. Ana, Pampanga served as the primary respondents of this study, representing individuals who require accessible and continuous health monitoring. Using purposive sampling, a non-probability method, the researchers exercised their judgment in selecting adult participants such as community members with chronic conditions (e.g., hypertension), local health workers, and IT experts who could best evaluate or benefit from the system. This deliberate selection ensured that feedback on the usability, affordability, and effectiveness of HealthX was both relevant and informed. A similar purposive sampling approach was employed by Cruz et al. (2022) in Community-Based IoT Health Solutions, wherein rural residents and health professionals were intentionally chosen to assess real-time patient monitoring technologies. In the present study, a total of 30 respondents was considered ideal, as this number provided a sufficient range of perspectives to generate meaningful and credible data while maintaining manageability and analytical depth. In purposive sampling, adequacy is determined not by sample size but by the richness and relevance of the information obtained; thus, 30 carefully selected participants were deemed appropriate to yield comprehensive insights into the practicality and applicability of the proposed system within Sta. Ana's healthcare context.

Table 1

Distribution of Respondents and Total Number of Participants

Category	Description	Number
Local Health Workers	Barangay health workers or nurses assessing medical relevance	10
Residents of Sta. Ana	Community members providing user-experience feedback	10
IT Experts	Professionals evaluating system architecture and technical reliability	10

Table 1 shows that a total of 30 participants were purposively selected, representing three key respondent groups—local health workers, residents of Sta. Ana, and IT experts. Each group contributed equally with ten participants, ensuring a balanced evaluation of the HealthX system in terms of medical relevance, user experience, and technical reliability.

Research Instrument

Survey questionnaires were distributed among the selected respondents, with the overall instrument reviewed and validated by a research expert to ensure clarity, relevance, and content validity. The survey was adopted from ISO/IEC 25010 and ISO/IEC 25012 standards to evaluate the functionality, efficiency, user satisfaction, and overall performance of the

HealthX. The expert's evaluation helped refine the questionnaire so that it accurately measures the intended variables and maintains high methodological rigor. The final questionnaire gathered quantitative data on key parameters such as system reliability, measurement accuracy of vital signs (SpO_2, heart rate, body temperature, blood pressure and BMI), data integrity, ease of use, and real-time connectivity through the ESP32 microcontroller. This expert-validated approach ensured a comprehensive assessment of the system's effectiveness, affordability, and suitability for continuous healthcare monitoring in remote communities.

Data Gathering

Data were gathered through field testing and paper-based structured surveys involving 30 purposively selected participants including residents, local healthcare workers, and IT experts in Sta. Ana, Pampanga chosen through the researchers' judgment for their ability to provide informed feedback. Prior to data collection, ethical standards were strictly observed in compliance with the Data Privacy Act of 2012 (R.A. 10173). The researchers explained the study's objectives and significance, and informed-consent forms were signed to ensure privacy, confidentiality, and voluntary participation. During field tests, the HealthX system recorded vital signs such

as SpO_2, heart rate, body temperature, blood pressure and BMI, which were simultaneously compared with certified medical devices to validate measurement accuracy. After testing, paper survey questionnaires adapted from ISO/IEC 25010 and ISO/IEC 25012 standards were distributed and completed by the same 30 respondents to assess functionality, reliability, usability, and data quality. Feedback highlighted measurement precision, real-time data transmission, ease of operation, and overall user satisfaction. All sensor readings and collected paper questionnaires were compiled, tabulated, and analyzed using descriptive statistics including mean, standard deviation, and correlation tests to evaluate the system's effectiveness and suitability for remote healthcare deployment.

Statistical Treatment

This study employed descriptive statistics to analyze user acceptance, expert evaluations, and overall system performance of HealthX in terms of efficiency, sustainability, and reliability, using a 4-point Likert scale to measure respondents' levels of agreement. Statistical tools such as mean, standard deviation, frequency, and percentage were utilized to summarize and interpret the collected data accurately. The mean determined the average perception of respondents, while the standard deviation reflected the degree of variation in their responses, indicating the level of consistency or disparity

in their evaluation of the system. A higher mean value represented a more favorable perception toward HealthX, whereas a higher standard deviation suggested differences in user experiences. Frequency and percentage distributions were used to identify response trends and determine whether the system was perceived as highly effective, moderately efficient, or in need of improvement. To assess the accuracy and reliability of the system, regression analysis was performed to examine the relationship between the sensor readings and the corresponding measurements from certified medical devices. The evaluation adhered to the ISO 25010 (Software Quality Evaluation) and ISO 25012 (Hardware Performance Evaluation) frameworks to ensure a standardized and structured assessment of system quality.

Table 2

Guideline Interval for Agreement Levels of Respondents

Point Scale	Mean	Descriptive Rating
4	3.01 – 4.00	Strongly Agree
3	2.01 – 3.00	Agree
2	1.01 – 2.00	Disagree
1	0.01 – 1.00	Strongly Disagree

Table 2 shows that the mean ranges indicate the respondents' levels of agreement based on a 4-point Likert scale, where higher mean values reflect stronger positive perceptions toward the HealthX system. A mean range of 3.01–4.00 signifies "Strongly Agree," showing that respondents expressed very favorable feedback. The range of 2.01–3.00 corresponds to "Agree," indicating general satisfaction, while 1.01–2.00 represents "Disagree," reflecting moderate disagreement. Lastly, 0.01–1.00 denotes "Strongly Disagree," suggesting minimal or negative perceptions. This scale serves as a guideline for interpreting survey results and evaluating the system's usability, reliability, and overall user satisfaction.

$$\overline{X} = \frac{\sum X}{N}$$

Where:

$\bar{x}$ = mean

x = data

N = number of data

System Development Life Cycle (SDLC)

The development of the HealthX: A Low-Cost IoT-Based Health Monitoring System follows the V-Model Software Development Life Cycle (SDLC), which emphasizes verification and validation at every stage. Each phase in the verification path corresponds directly to a testing activity in the validation path, ensuring high reliability, data accuracy, and user satisfaction.

Figure 2

V-Model Software Development Life Cycle (SDLC)

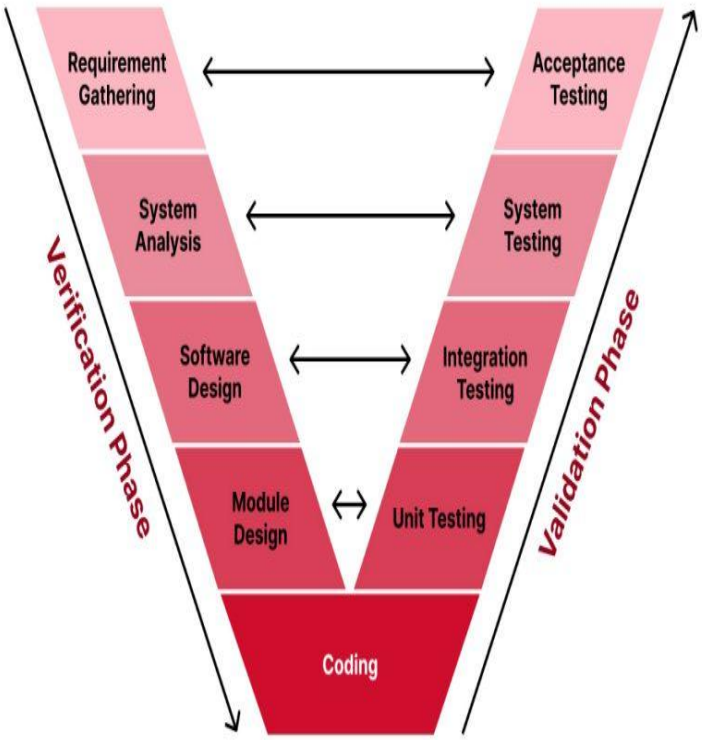

Figure 2 shows that the adopted from "V-Model in Software Development Life Cycle" by Guru99, 2024, https://www.guru99.com/v-model-software-testing.html. The diagram illustrates the dual-phase approach of the V-Model used in developing HealthX, emphasizing verification and validation throughout the process.

Planning

The HealthX was developed using a V-Model approach to ensure systematic development and thorough validation at each stage. The Planning Phase began in Week 1 to Week 4 of September, during which the project scope, objectives, and feasibility were defined. The team identified the required hardware—ESP32 board, MAX30102 SpO_2 and heart-rate sensor, DS18B20 body-temperature sensor, Load Cell with HX711, and a blood-pressure sensor module—and the software stack, including Arduino IDE (C++), Flutter for the mobile interface, and PHP/MySQL for the web dashboard. Stakeholder consultations with Sta. Ana residents and local healthcare workers helped refine requirements and establish success criteria. The Design Phase, running from Week 1 to Week 4 of October, produced detailed circuit schematics, database structures, and system architecture diagrams. This phase ensured seamless integration of all sensors with the ESP32 microcontroller and defined the user flows for both the Flutter mobile

application and the PHP/MySQL web dashboard. During the Development Phase from Week 1 of November to Week 4 of December, the hardware prototype was assembled and firmware for real-time data acquisition was coded in C++. Parallel sprints focused on building the Flutter mobile app for patient access and the web-based dashboard for healthcare professionals. Sensor calibration and connectivity were iteratively tested in each sprint to maintain accuracy for vital parameters—SpO_2 (normal range 95–100%), heart rate (60–100 bpm), body temperature (36.5–37.5 °C), and blood pressure (≈120/80 mmHg). The Testing Phase, held from Week 1 to Week 4 of January, validated system reliability and data accuracy by comparing sensor outputs with certified medical devices. Quality assessments followed ISO/IEC 25010 for software and ISO/IEC 25012 for hardware/data quality, focusing on functionality, usability, and data integrity. Following successful tests, the Deployment Phase spanned Week 1 to Week 3 of February, introducing the system to selected households and healthcare workers in Sta. Ana, Pampanga, for real-world trials. User training sessions were provided to ensure proper operation and to gather experiential feedback. Finally, the Evaluation and Review Phase, conducted in Week 3 to Week 4 of February, analyzed survey responses and field data using descriptive statistics—mean,

standard deviation, frequency, and percentage distributions—to assess performance, user satisfaction, and overall system effectiveness.

Requirement Analysis

During this phase, the project objectives, scope, and feasibility were defined. Consultations were conducted with residents and healthcare workers in Sta. Ana, Pampanga to identify local health monitoring needs. The system requirements were established, focusing on six vital health indicators—oxygen saturation (SpO_2), heart rate, body temperature, blood pressure, weight, and BMI. The required hardware components, including the ESP32 microcontroller, MAX30102 SpO_2 and heart-rate sensor, DS18B20 body temperature sensor, HX711 load cell module, and blood pressure sensor, were identified. Software requirements were also outlined, specifying Arduino IDE (C++), Flutter for the mobile interface, and PHP/MySQL for the web dashboard.

System Design

This phase produced the system's architectural design and data flow structure. Circuit schematics, database schema, and overall system architecture were documented to ensure seamless sensor integration with the ESP32 microcontroller. The design also included user interface flow

diagrams for both the mobile application and the web-based dashboard, ensuring intuitive access for patients and healthcare professionals.

Figure 3

Block Diagram Model of the HealthX

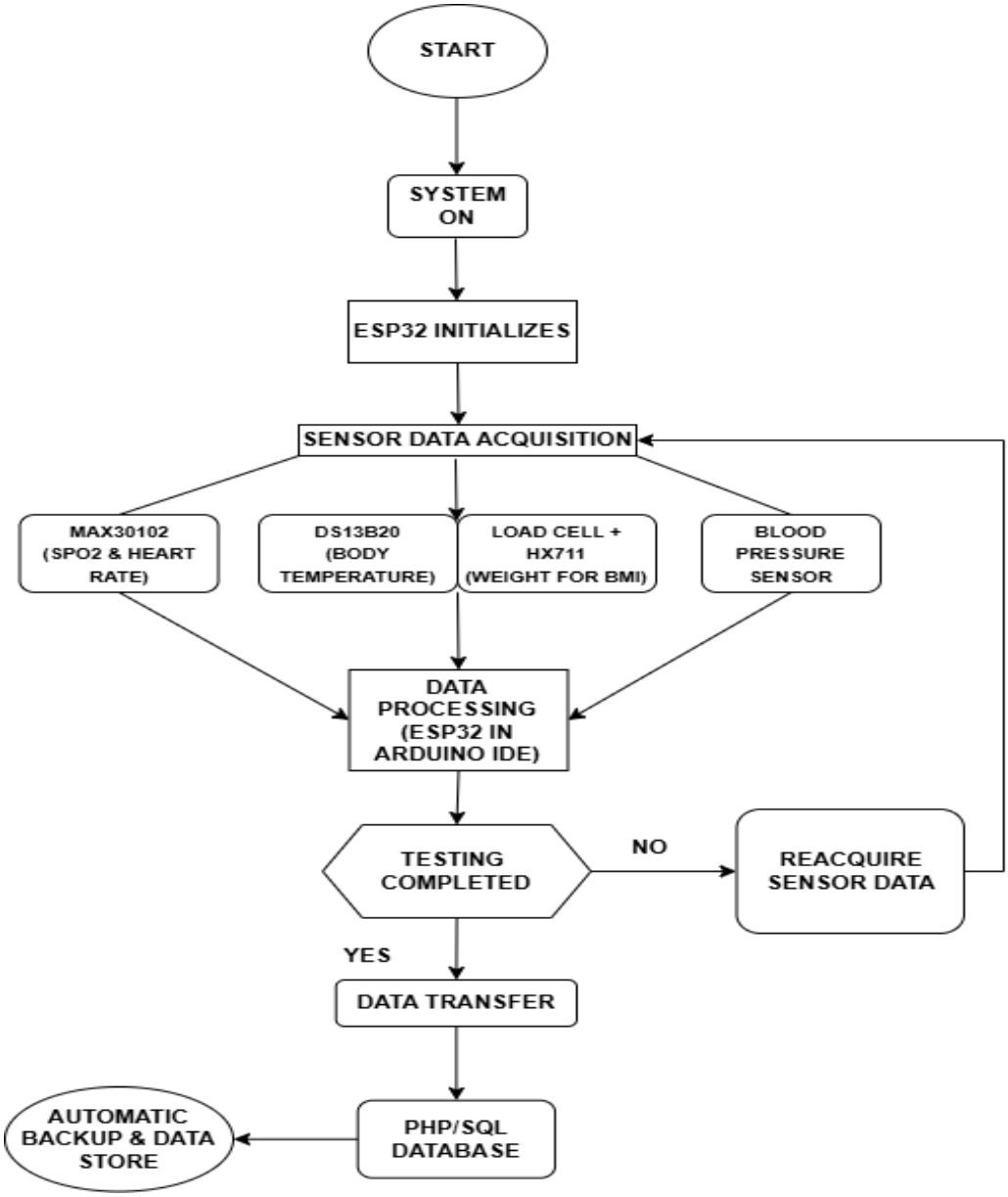

Figure 3 shows that the diagram presents the detailed system architecture and workflow of the HealthX Low-Cost IoT-Based Health Monitoring System, which operates through an ESP32 Wi-Fi microcontroller. The ESP32 serves as the central processing unit that collects physiological data from multiple sensors: the MAX30102 for measuring oxygen saturation (SpO_2) and heart rate, the DS18B20 for body temperature detection, the HX711 load cell module for weight measurement and BMI computation, and a blood pressure sensor for systolic and diastolic readings. These sensor readings are processed by the ESP32, which analyzes and formats the data before transmitting it to a PHP/MySQL web server for storage and doctor access. Simultaneously, the data are displayed on the Flutter-based mobile application for real-time visualization, allowing both patients and healthcare professionals to monitor vital signs anytime, thereby enhancing accessibility, accuracy, and continuity of healthcare monitoring.

Implementation Phase

During the Implementation Phase, the hardware and software components of the HealthX system were fully developed and integrated. The ESP32 microcontroller was programmed using C++ in the Arduino IDE to interface with biomedical sensors such as the MAX30102 (SpO_2 and heart rate), DS18B20 (body temperature), Load Cell with HX711 amplifier (BMI), and a blood pressure sensor module. The Flutter mobile application was designed to display real-time vital readings, while the PHP/MySQL web dashboard enabled doctors to monitor and manage patient data efficiently. Each module—data acquisition, processing, database communication, and user interface—was systematically configured and tested to ensure proper synchronization, responsiveness, and reliable performance across devices.

Testing and Validation Phase

This phase involved comprehensive testing to verify the system's accuracy, reliability, and stability. Each sensor was calibrated and compared with certified medical-grade instruments to validate measurements such as SpO_2 (95–100%), heart rate (60–100 bpm), body temperature (36.5–37.5 °C), and blood pressure (≈120/80 mmHg). Connectivity assessments confirmed stable real-time data transmission between the ESP32, Flutter mobile app, and PHP/MySQL web server. Usability tests ensured both healthcare providers

and users could easily interpret the health data. Quality assurance was guided by ISO/IEC 25010 for software performance and ISO/IEC 25012 for data accuracy and integrity.

Deployment Phase

After successful validation, the system was deployed at a community health center in Sta. Ana, Pampanga, for pilot implementation. Healthcare workers and selected residents were trained in operating the device, viewing results through the mobile app, and troubleshooting basic issues. The deployment enabled real-world testing of the system's functionality, data accuracy, and user experience in an actual healthcare environment. Continuous data logging during the pilot phase provided additional insights into performance consistency under daily use.

Evaluation and Continuous Improvement

The Evaluation and Maintenance Phase focused on analyzing collected data and ensuring continuous improvement through ongoing testing and recalibration. Statistical tools such as mean, standard deviation, and frequency distribution were used to assess measurement reliability and user satisfaction, while regression analysis examined correlations between environmental conditions and sensor accuracy. Continuous testing was conducted throughout this phase to monitor long-term performance and detect

potential drifts in sensor readings. Feedback from healthcare workers and users guided both software updates and hardware refinements, improving stability, usability, and accuracy. This phase ensured that the HealthX system remained reliable, well-maintained, and ready for scalable deployment in broader community healthcare settings.

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